

# **SUMMARY**

## **1. Best Performing Model**

The XGBoost model performed best among all evaluated models.

Reasons:

- Highest fraud detection capability on imbalanced data.
  - Better Precision-Recall performance compared to baseline models.
  - Handles non-linear fraud patterns effectively.
  - Strong generalization with engineered features.
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## **2. Why PR-AUC matters more than Accuracy in Fraud Detection**

Fraud datasets are highly imbalanced, where fraudulent transactions represent only a very small percentage of total transactions.

Therefore:

- A model can achieve very high accuracy by predicting all transactions as non-fraud.
- This would fail to detect actual fraud cases.

Precision-Recall AUC is more important because:

- Precision measures how many flagged frauds are truly fraud.
  - Recall measures how many actual frauds are detected.
  - PR-AUC provides a better evaluation for rare-event classification.
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## **3. Top 3 Fraud Signals identified by SHAP**

Based on SHAP analysis, the strongest fraud indicators were:

### **1. TransactionAmt**

- Extremely high transaction values increased fraud probability.

### **2. HourOfDay**

- Late-night and unusual transaction times showed higher fraud risk.

### **3. AmtToMeanRatio**

- Transactions significantly above normal customer spending patterns were suspicious.
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## **4. Common Characteristics of Critical Risk Transactions**

Critical-risk transactions generally showed:

- High transaction amount
  - Unusual transaction time
  - Abnormal spending ratio
  - Higher predicted fraud probability
  - Multiple feature anomalies simultaneously
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## 5. Two Actionable Fraud Prevention Policies

### Policy 1: Real-time transaction blocking

Automatically block transactions when fraud probability exceeds the optimal threshold.

Benefit:

- Prevents financial loss immediately.

### Policy 2: Multi-factor verification

Require OTP or additional authentication for medium/high-risk transactions.

Benefit:

- Reduces false positives while improving security.
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## 6. Estimated Money Saved Annually

Assuming:

- Average fraud amount = detected average fraud amount
- Model detects majority of fraud cases early

Estimated annual savings:

Potentially millions in prevented fraudulent losses, depending on transaction volume.

This can significantly reduce operational and reimbursement costs.

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## 7. Model Limitations

Current limitations include:

- Limited feature availability
- No customer behavioral history
- Possible concept drift over time

- May produce false positives
  - Requires periodic retraining
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## **8. Additional Data That Could Improve Performance**

Future improvements can include:

- Customer historical behavior
- Device fingerprint data
- IP address and geolocation
- Merchant risk scores
- Browser/device metadata
- Velocity-based features (multiple rapid transactions)

These additional signals can improve fraud detection accuracy further.